

Unit-2

- **The scope of learning, popular architectures**
- **An overview of Parallel and Distributed Processes (PDP),**
- **linear associative models, and stochastic networks**

1. Scope of Learning

(A) What is Learning?

A system **learns** if its performance improves with experience.

According to Tom M. Mitchell:

“A computer program is said to learn from experience E with respect to some task T and performance measure P if its performance at T, as measured by P, improves with experience E.”

Scope of Learning (AI/ML)

The scope of learning in AI is vast and constantly evolving, enabling machines to perform tasks that previously required human intelligence. Instead of relying on explicitly programmed, fixed rules, systems use data-driven models to adapt and make decisions in dynamic environments.

Key aspects of the scope include:

- **Automation:** Performing repetitive tasks across industries like manufacturing and data entry.
- **Data Analysis and Prediction:** Processing large datasets to uncover patterns, trends, and actionable insights for business decisions, fraud detection, and demand forecasting.
- **Natural Language Processing (NLP):** Understanding and generating human language for applications such as chatbots, language translation, and sentiment analysis.
- **Computer Vision:** Interpreting and processing visual data for image recognition, medical image analysis, and autonomous navigation.
- **Robotics:** Powering robots to perform tasks requiring precision and adaptability in environments like surgical procedures and autonomous vehicles.
- **Personalization:** Customizing user experiences and services, such as product recommendations on e-commerce sites or personalized learning paths in education.

(B) Types of Learning (Scope)

1 Supervised Learning

- Data with labels
- Used for classification & regression
- Example: Disease classification in chest X-rays

Algorithms:

- Linear Regression
- Logistic Regression
- SVM
- KNN
- Neural Networks

2 Unsupervised Learning

- No labeled data
- Discover hidden patterns

Algorithms:

- K-means
- Hierarchical Clustering
- PCA
- Autoencoders

3 Semi-Supervised Learning

- Few labeled + many unlabeled samples
- Useful in medical imaging (limited annotations)

4 Reinforcement Learning

- Agent interacts with environment
- Learns via reward system
- Used in robotics, game AI

Example: AlphaGo defeating Lee Sedol.

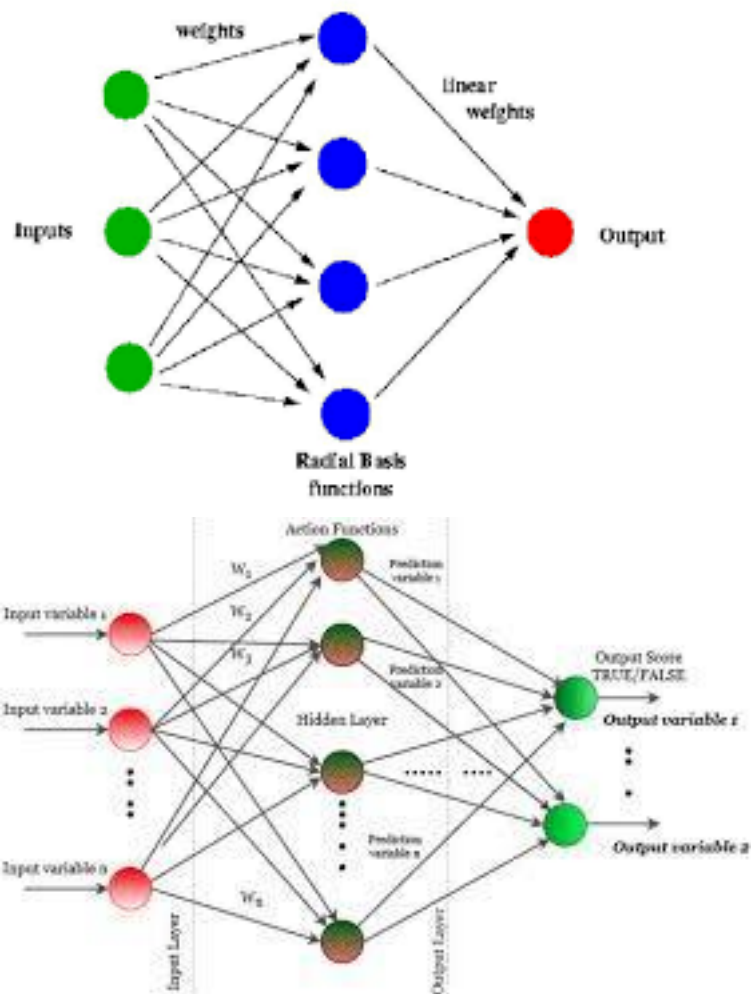
(C) Scope of ML in Real-World Applications

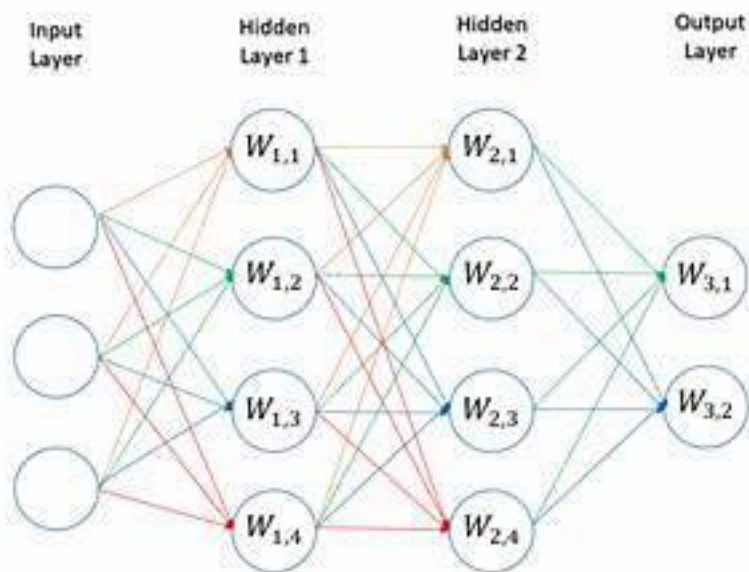
Domain	Applications
Healthcare	Cancer detection, COVID diagnosis
Computer Vision	Face recognition, Object detection
NLP	Chatbots, Translation
Finance	Fraud detection
Autonomous Systems	Self-driving cars

♦ 2. Popular Deep Learning Architectures

Now focusing on architectures (very important for CNN-based research like yours).

1 Artificial Neural Network (ANN)



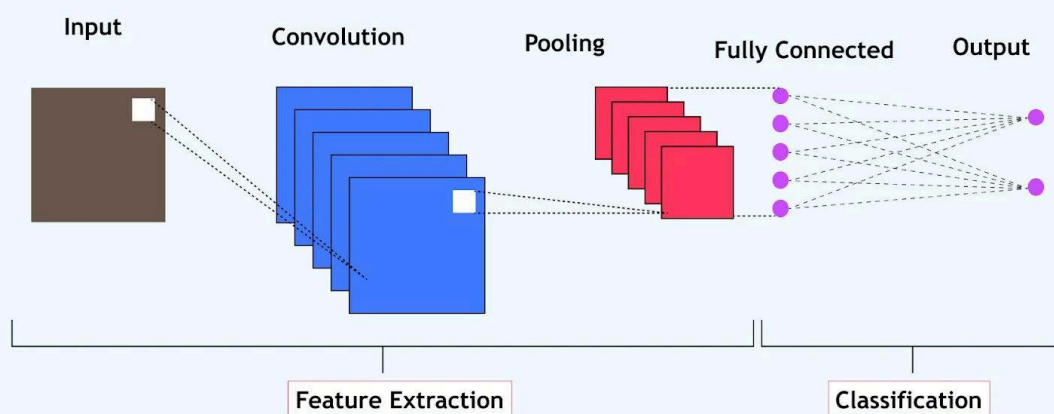


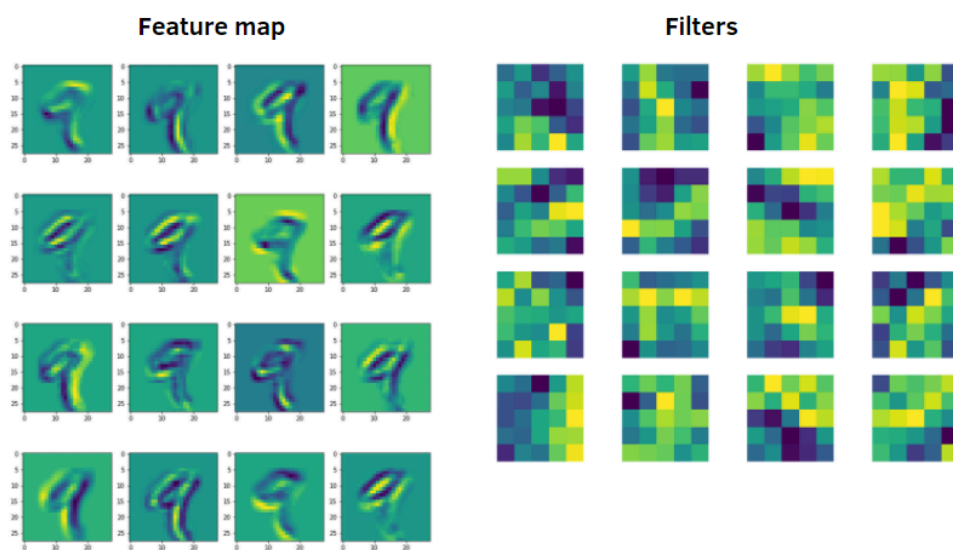
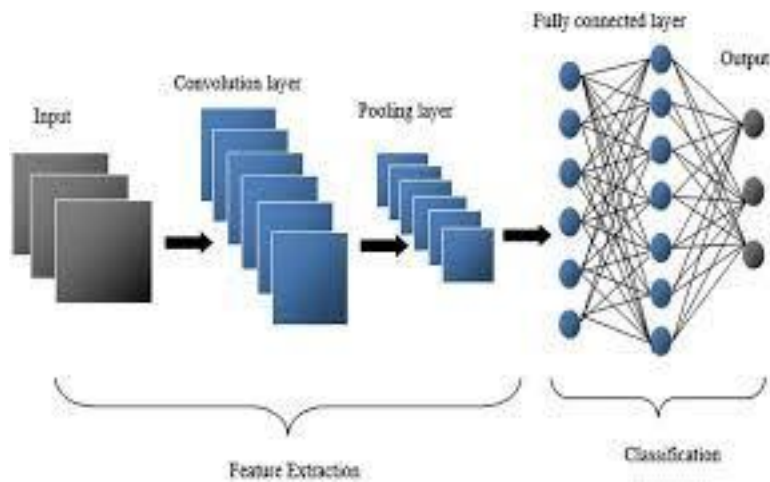
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- Input layer
- Hidden layers
- Output layer
- Fully connected layers
- Used for tabular data, regression

2 Convolutional Neural Network (CNN)

The Architecture of Convolutional Neural Networks





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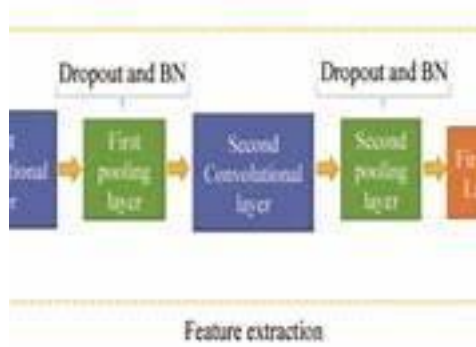
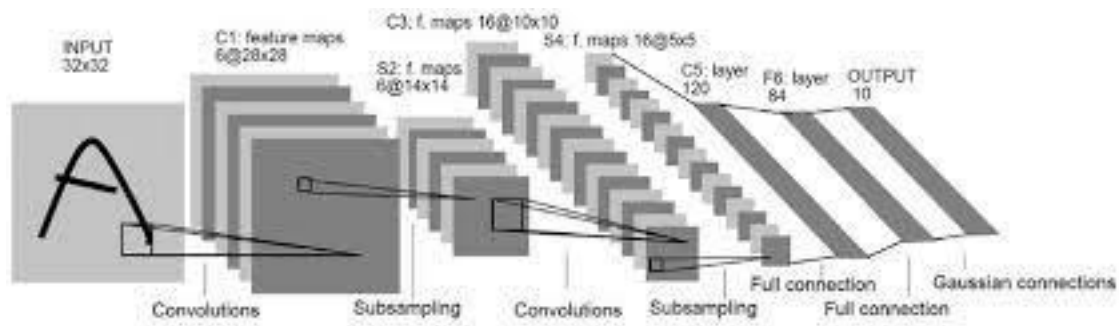
- Convolution layers
- Pooling layers
- Fully connected layers
- Best for image processing

Used in:

- Medical imaging
- Object detection
- Face recognition

Popular CNN Architectures

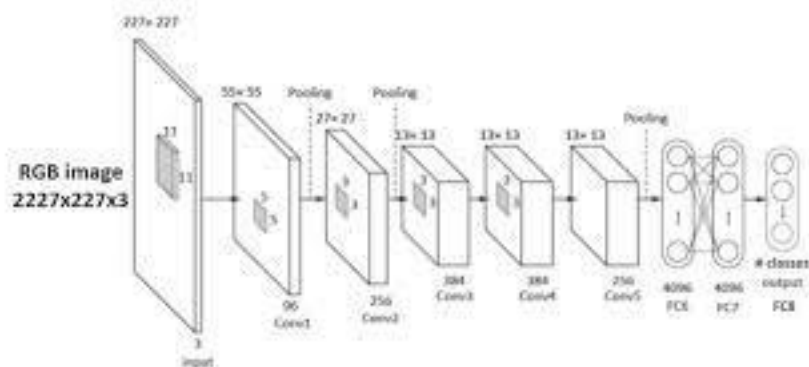
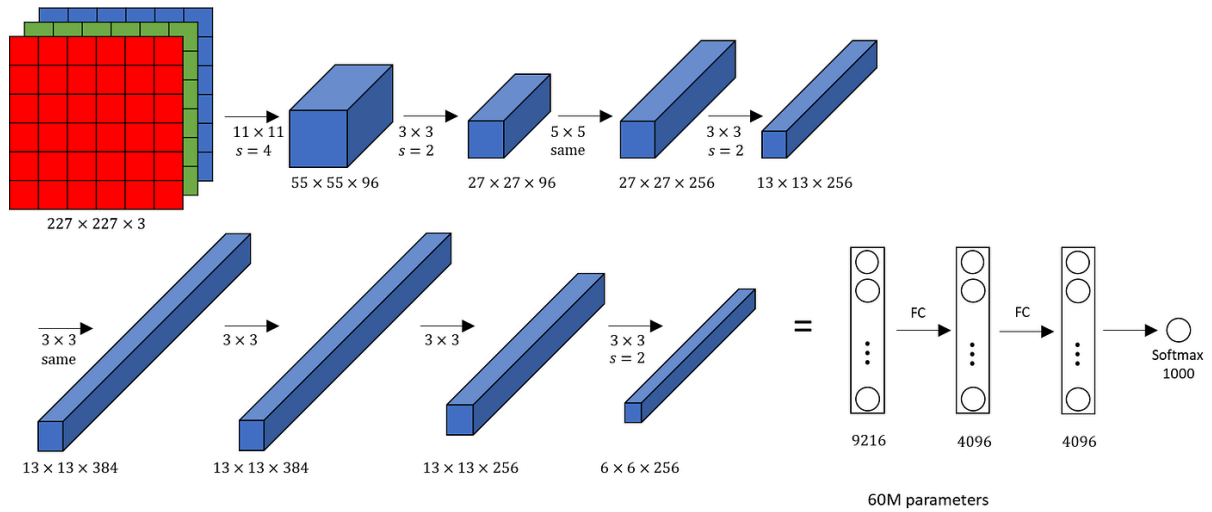
- ♦ LeNet

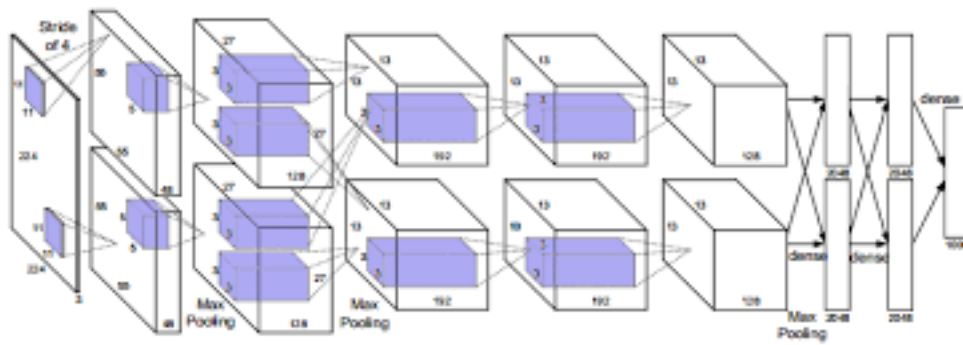


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- Early CNN (1998)
- Handwritten digit recognition

◆ AlexNet

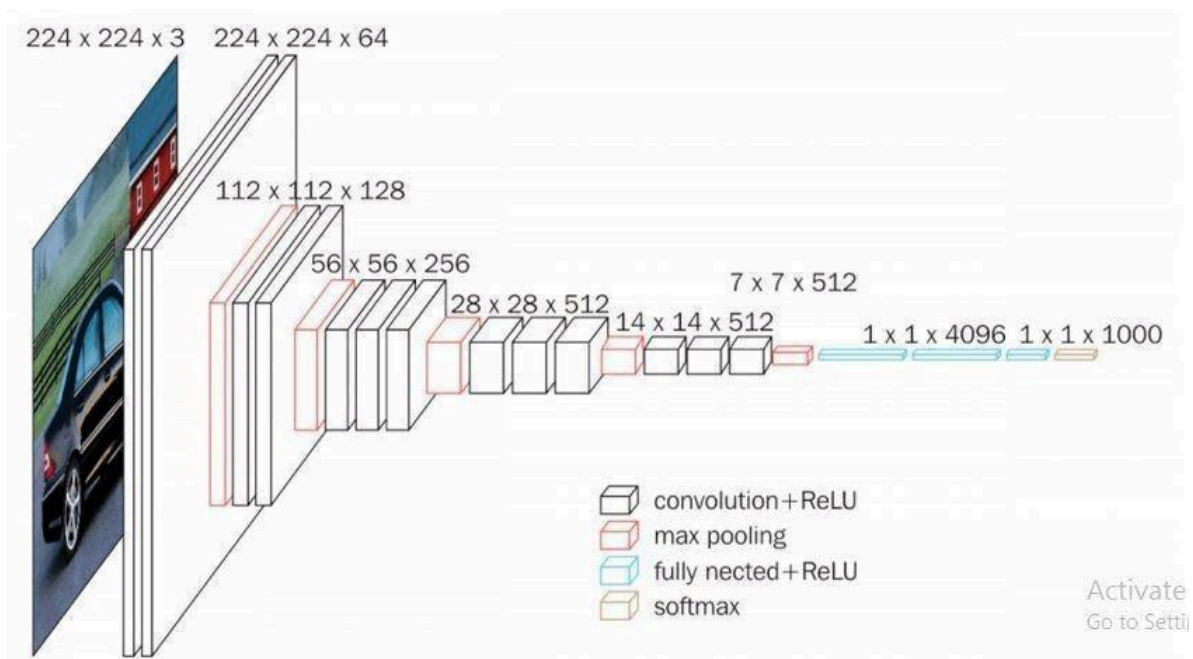


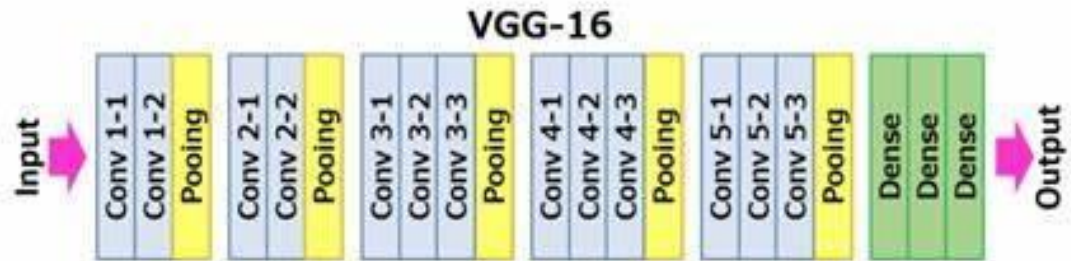


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- Won ImageNet 2012
- Introduced ReLU, Dropout

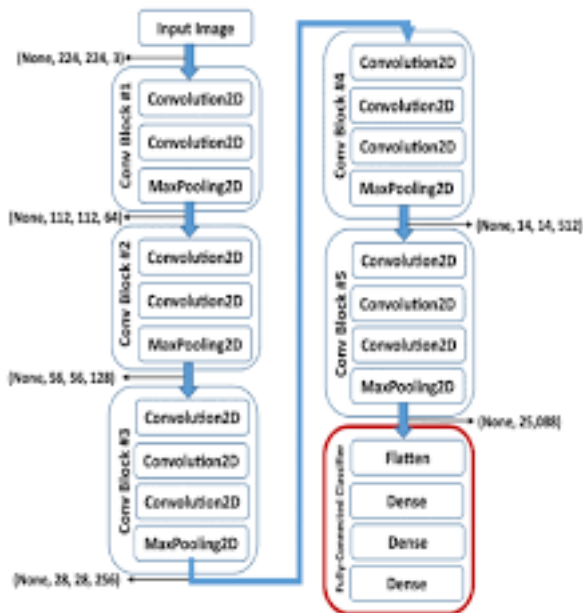
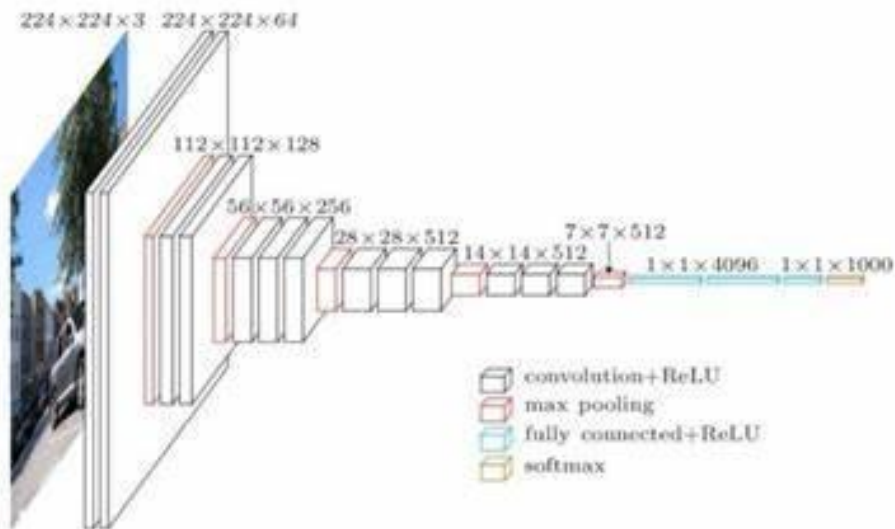
◆ VGG-16





The Architecture

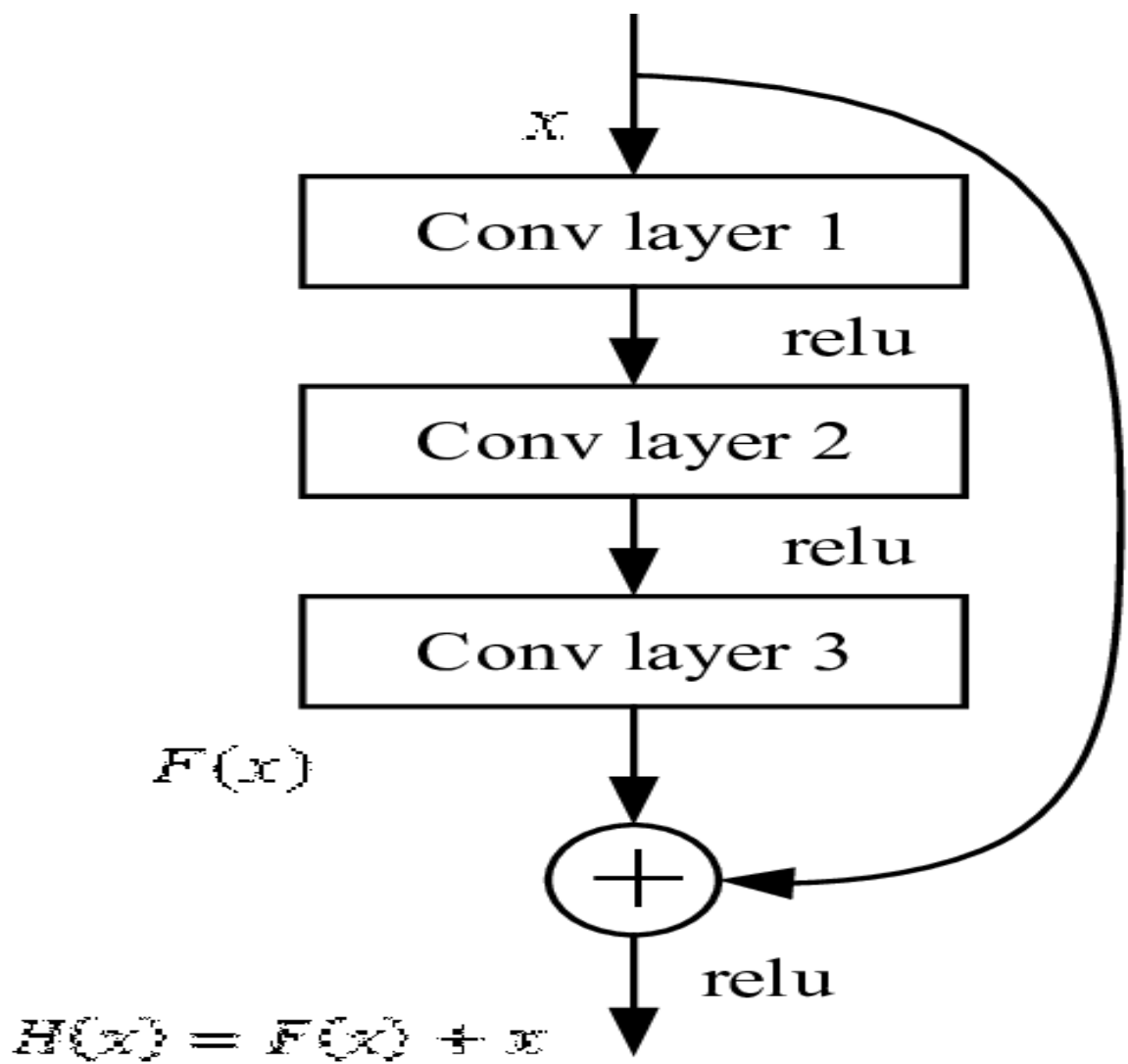
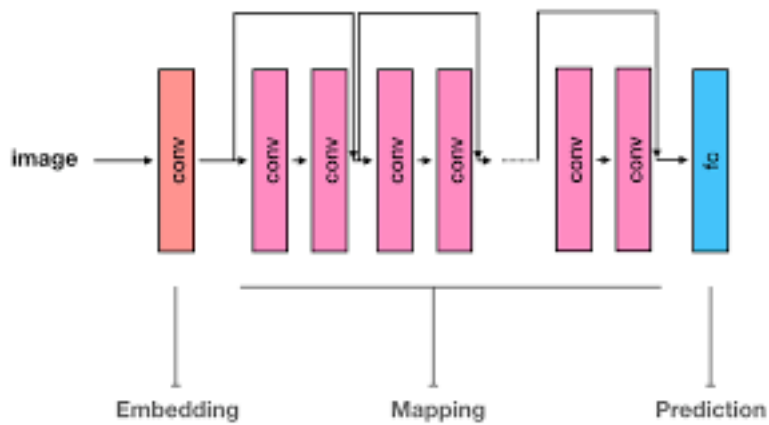
The architecture depicted below is VGG16.

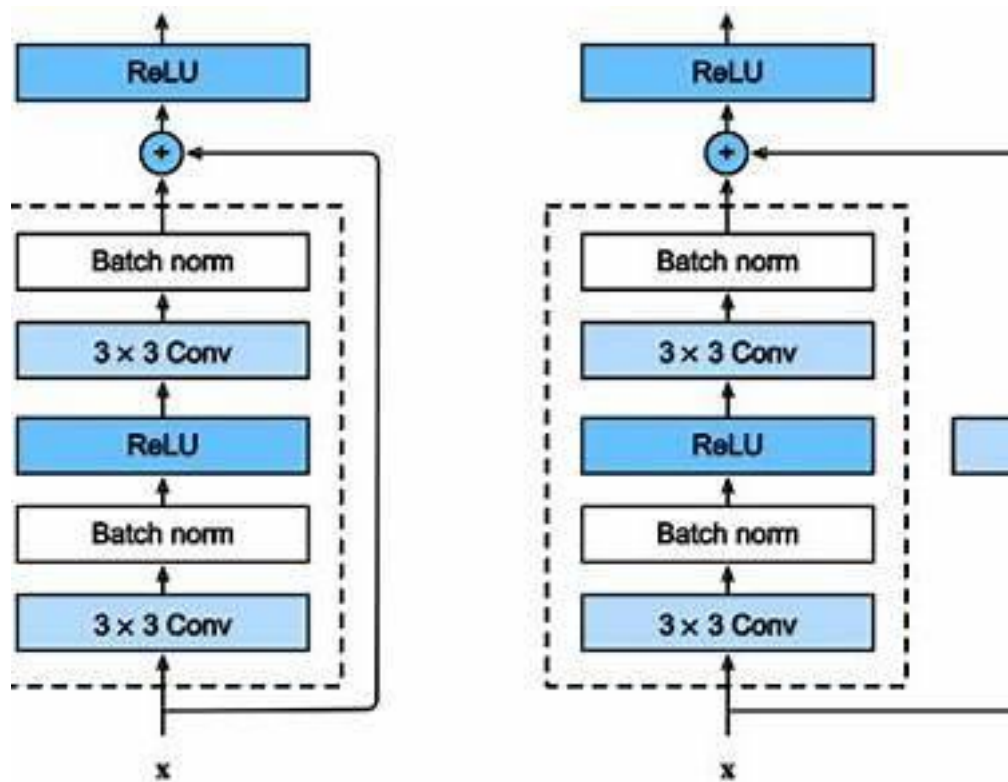


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- Very deep (16 layers)
- Uses 3x3 filters
- Heavy parameters

◆ ResNet

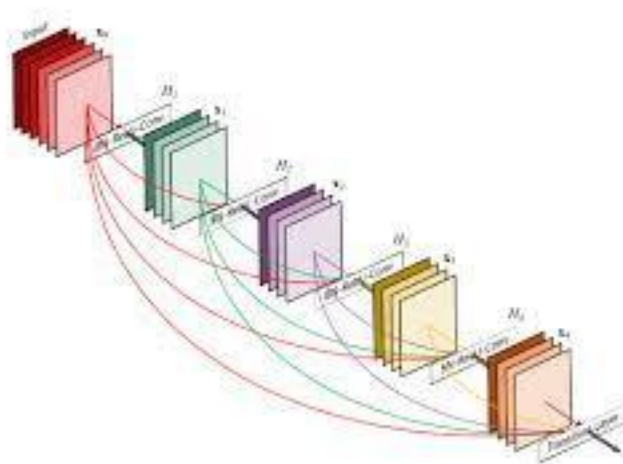


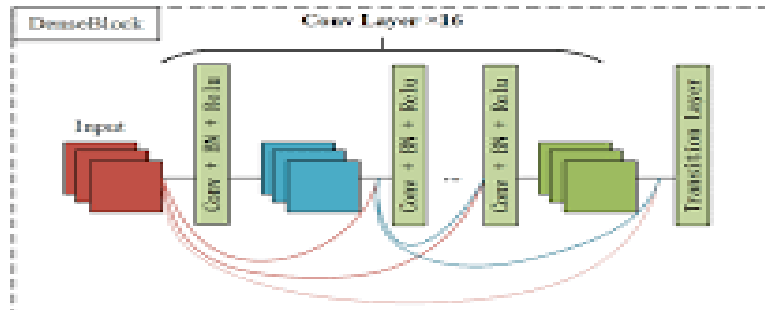
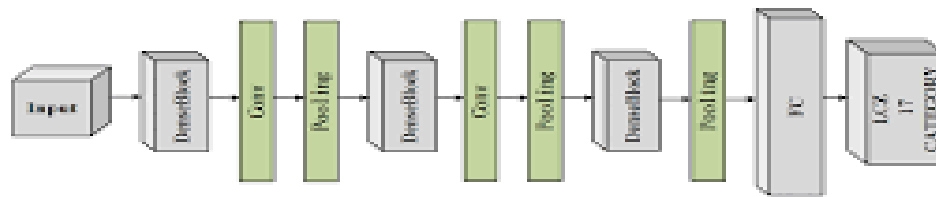


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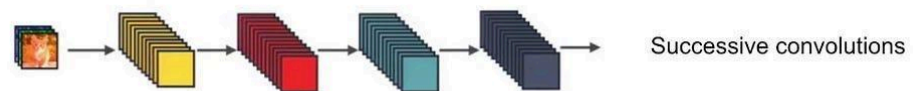
- Introduced Skip Connections
- Solves vanishing gradient
- Versions: ResNet-50, ResNet-101

◆ DenseNet

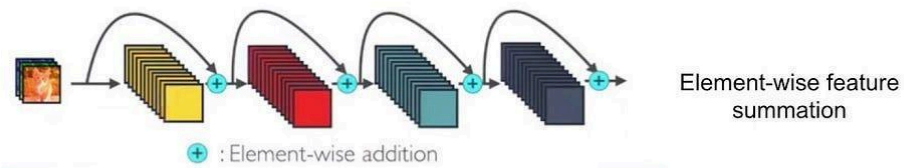




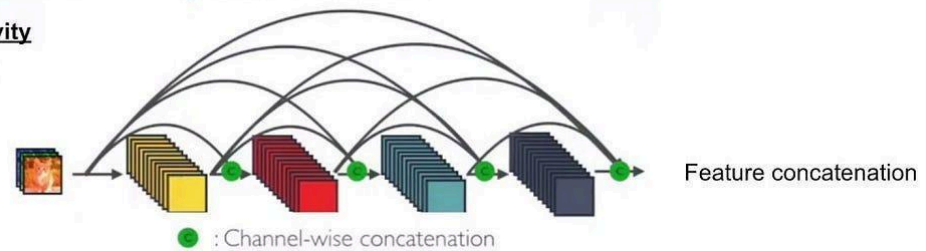
Standard Connectivity



Resnet Connectivity



DenseNet Connectivity



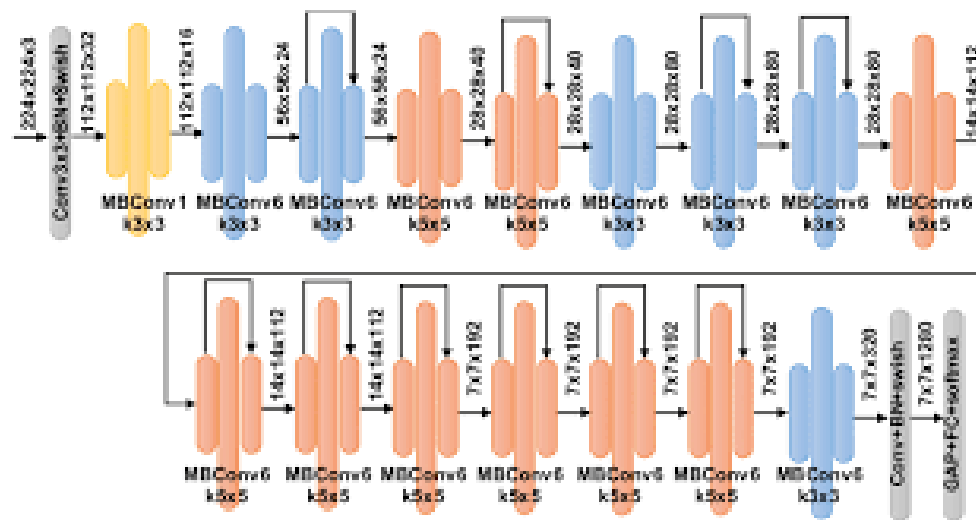
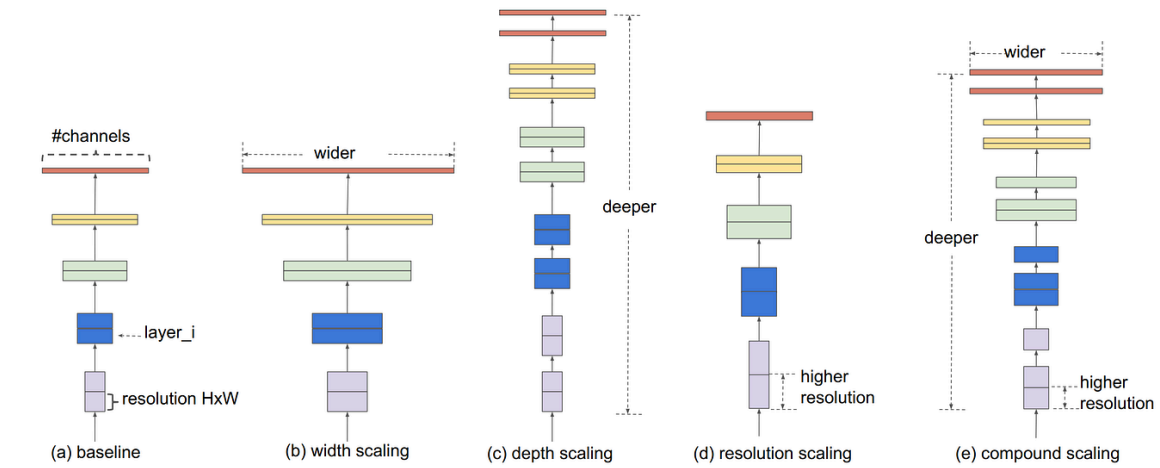
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- Dense connections
- Better feature reuse
- Fewer parameters than ResNet

◆ EfficientNet

EfficientNet Architecture

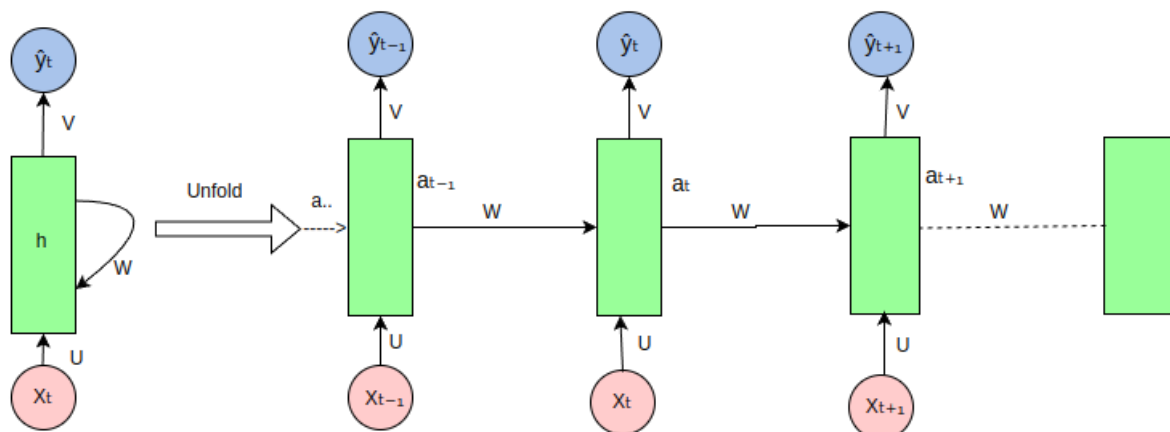




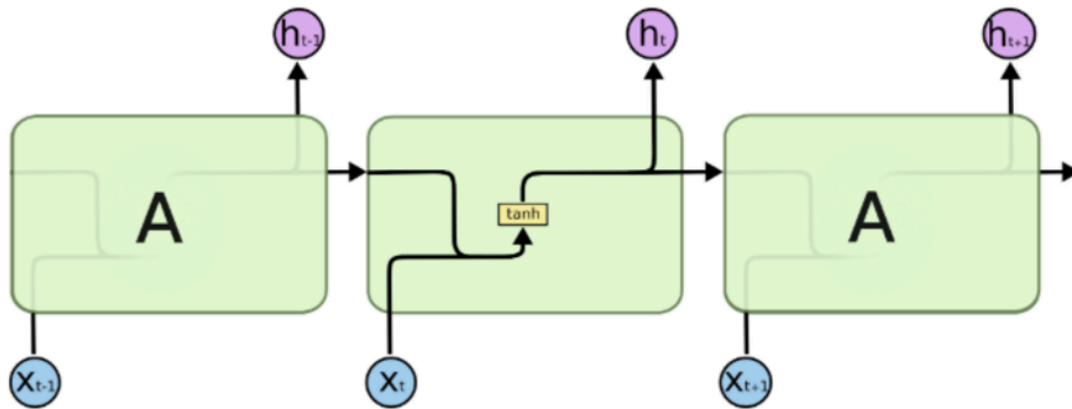
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- Compound scaling (depth + width + resolution)
- State-of-the-art performance

3 Recurrent Neural Network (RNN)



RNN



The repeating module in a standard RNN contains a single layer.

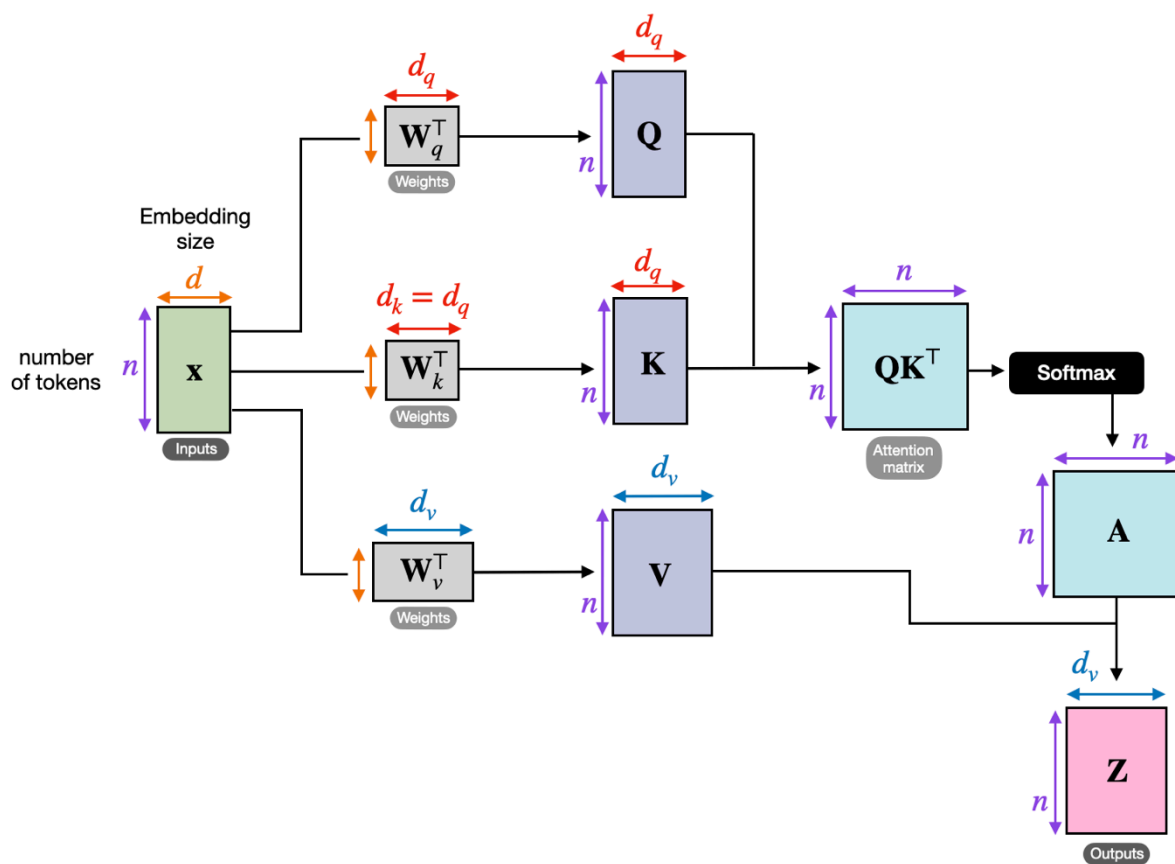
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- Sequential data
- NLP, time series

Advanced versions:

- LSTM
- GRU

4 Transformer Architecture

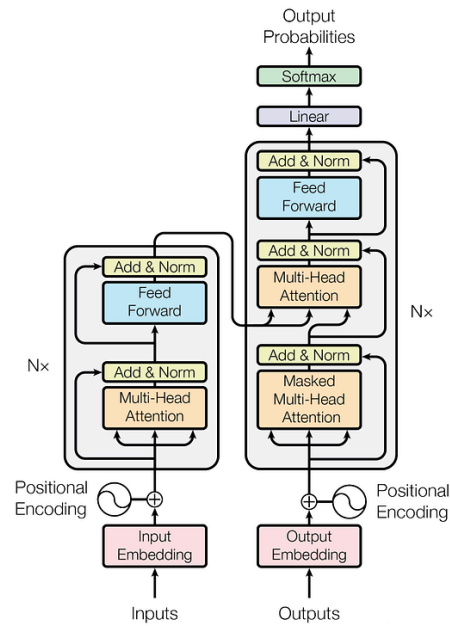


BERT

Encoder

GPT

Decoder



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Based on attention mechanism.

Used in:

- NLP
- Vision Transformers (ViT)

Example: Vision Transformer

♦ Summary Table

Architecture	Best For	Key Idea
ANN	Tabular Data	Fully connected layers
CNN	Images	Convolution filters
RNN	Sequence	Memory through recurrence
ResNet	Deep CNN	Skip connections
DenseNet	Efficient CNN	Dense connectivity
EfficientNet	Scalable CNN	Compound scaling
Transformer	NLP & Vision	Self-attention